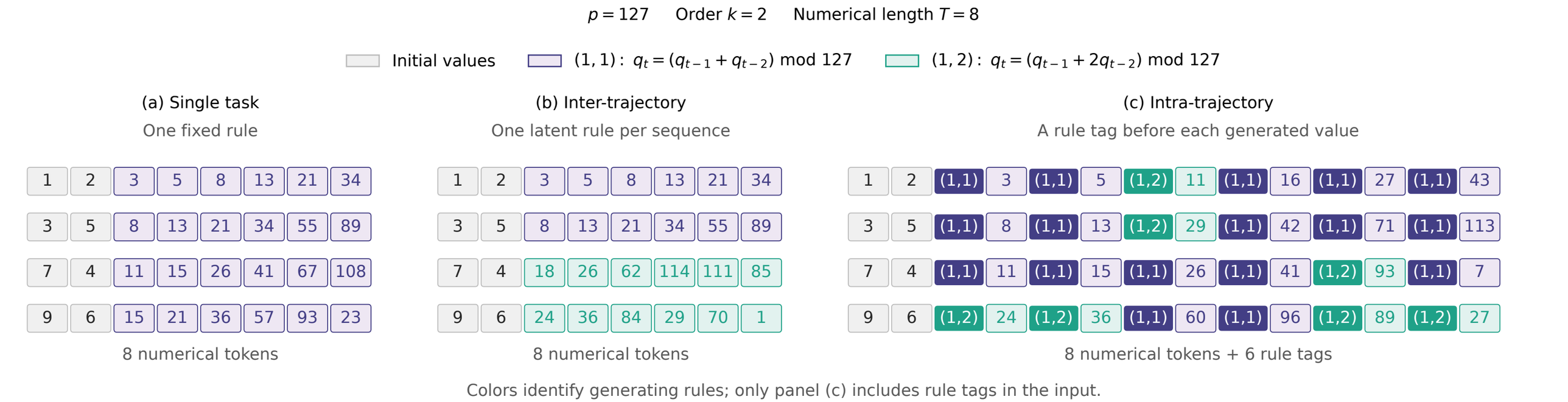
$p = 127$ Order $k = 2$ Numerical length $T = 8$
Initial values
$(1,1)$: $q_t = (q_{t-1} + q_{t-2}) \bmod 127$
$(1,2)$: $q_t = (q_{t-1} + 2q_{t-2}) \bmod 127$
(a) Single task
One fixed rule
1 2 3 5 8 13 21 34
3 5 8 13 21 34 55 89
7 4 11 15 26 41 67 108
9 6 15 21 36 57 93 23
8 numerical tokens
(b) Inter-trajectory
One latent rule per sequence
1 2 3 5 8 13 21 34
3 5 8 13 21 34 55 89
7 4 18 26 62 114 111 85
9 6 24 36 84 29 70 1
8 numerical tokens
(c) Intra-trajectory
A rule tag before each generated value
1 2 (1,1) 3 (1,1) 5 (1,2) 11 (1,1) 16 (1,1) 27 (1,1) 43
3 5 (1,1) 8 (1,1) 13 (1,2) 29 (1,1) 42 (1,1) 71 (1,1) 113
7 4 (1,1) 11 (1,1) 15 (1,1) 26 (1,1) 41 (1,2) 93 (1,1) 7
9 6 (1,2) 24 (1,2) 36 (1,1) 60 (1,1) 96 (1,2) 89 (1,2) 27
8 numerical tokens + 6 rule tags
Colors identify generating rules; only panel (c) includes rule tags in the input.